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ABSTRACT

With the widespread integration of high-penetration renewable energy into distribution networks, the volatility and uncertainty of system operation have increased significantly, posing substantial challenges to traditional reactive voltage-power optimization. Existing methods based on mathematical programming or intelligent optimization generally suffer from low computational efficiency and a tendency to converge to local optima when dealing with high-dimensional non-convex optimization problems. Digital Twin (DT) technology offers a new approach for real-time grid perception and dynamic simulation; however, its decision-making core still relies on classical algorithms, which limit its analytical capability. Quantum Computing (QC), due to its inherent parallelism, provides revolutionary computational potential for solving complex combinatorial optimization problems. We innovatively integrate QC and DT technologies to construct a Quantum Computing-enhanced Digital Twin System (DTS) (QC-enhanced DTS). By establishing a high-fidelity virtual mapping and designing a hybrid quantum-classical algorithm, the system enables advanced predictive inference and cooperative optimization of distribution network operation. Comparative simulation experiments verify that the proposed method outperforms conventional algorithms by more than 24.3% in both convergence speed and global optimization capability, which effectively reduces system network losses and enhances voltage stability. Therefore, this research not only presents a significant exploration into the practical application of QC but also provides a novel direction for building highly resilient smart grids, offering considerable theoretical value and engineering application prospects.

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NOMENCLATURE

CPLEX	constraint programming LEX
DERs	distributed energy resources
DT	digital twin
HQCA	hybrid quantum-classical algorithm
MINLP	mixed-integer nonlinear programming
MIP	mixed-integer programming
NISQ	noisy intermediate-scale quantum
OLTC	on-load tap changer
PLA	piecewise linear approximation
PSO	particle swarm optimization
QC	quantum computing
QC-enhanced DTS	quantum computing-enhanced digital twin system
QUBO	quadratic unconstrained binary optimization
SVC	static var compensator
SOCR	second-order cone relaxation

I. INTRODUCTION

The global transition of the energy structure continues to deepen, leading to increased penetration of distributed photovoltaic (PV) and wind power in distribution networks.[1] The inherent stochastic and intermittent nature of renewable energy generation exacerbates power fluctuations, making reactive power

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balance and voltage stability control in distribution systems increasingly prominent.[2] Conventional local control strategies based on localized measurements are inadequate for achieving global coordination and optimization, while centralized optimization methods face computational complexity issues due to large-scale nonlinear programming.[3] Currently, the Reactive Voltage-Power Optimization (RVPO) problem is commonly formulated as a non-convex nonlinear programming model with multiple constraints. This problem is Non-Deterministic Polynomial time-hard (NP-hard), and its computational complexity grows exponentially with the system scale.[4] Although traditional optimization methods such as the interior-point method and particle swarm optimization (PSO) have been widely applied, they exhibit inherent limitations in real-time performance, global convergence, and handling uncertainties.[5] Existing research has yet to effectively resolve the inherent trade-off between computational efficiency and solution accuracy, and a breakthrough computational paradigm that simultaneously satisfies both real-time operation and optimality remains lacking.[6]

Amaral *et al.* proposed a second-order cone programming-based model for optimal distributed generation allocation and designed a relaxation conversion method; however, it suffers from insufficient relaxation precision and difficulties in guaranteeing global optimality.[7] In Ref. 8, a multi-time-scale voltage control architecture was introduced under a model predictive control framework, but it entails a heavy computational burden in rolling optimization and is challenging to deploy online. Peng and Delage developed a data-driven robust optimization strategy with a distributed decision-making mechanism, yet it exhibits excessive conservativeness and reduced economic efficiency.[9] Gholizadeh and Musilek proposed a deep reinforcement learning-based method for reactive power control using an end-to-end decision network, which requires extensive training samples and lacks interpretability.[10] Mylona and Bouhouras systematically elaborated an application framework of digital twin (DT) technology in power grids and designed a cyber-physical fusion system, but it lacks in-depth integration with advanced computational techniques.[11] Main *et al.* explored the theoretical prospects of quantum computing (QC) in power system optimization and proposed a general problem mapping methodology.[12] However, it does not provide practical hybrid algorithm designs for engineering applications. Díez-Valle *et al.* investigated the application of the Quantum Approximate Optimization Algorithm (QAOA) in microgrid economic dispatch and designed a quantum-classical hybrid workflow.[13] However, it does not account for complex distribution network topologies and multiple constraints. Lebedev *et al.* developed a cloud-platform-based demonstration of DT for distribution systems, which achieves multi-source data integration and visualization;[14] however, its analytical module still relies on conventional solvers with limited decision-making intelligence.

To address the computational bottlenecks of existing RVPO methods in handling high-dimensional non-convex problems and the insufficient decision-making capabilities of DT systems,[15] this study constructs a reactive power-voltage collaborative optimization framework deeply integrated with QC-enhanced Digital Twin System (DTS). First, a high-fidelity DT model of the distribution network is established to achieve accurate virtual mapping and real-time data interaction with the physical system. Then, the conventional RVPO problem is transformed into a Quadratic Unconstrained Binary Optimization (QUBO) model amenable to quantum processing. On this basis, a hierarchically collaborative hybrid quantum-classical algorithm (HQCA) is designed, which incorporates pre-processing and post-processing stages, and invokes the QC-enhanced DTS to solve the core optimization problem. Through closed-loop interaction between the DT platform and the quantum algorithm, the proposed framework ultimately enables proactive inference of future system states and optimal decision-making. It is anticipated that this method will significantly enhance the speed and accuracy of solving optimization problems, providing key technical support for the secure, stable, and economic operation of distribution networks with high penetration of renewable energy.

The main technical innovations of this study lie in the following aspects:

(1) A deeply integrated system architecture combining quantum computing (QC) and digital twin (DT) is proposed. This framework not only employs DT for holographic perception and dynamic simulation of physical systems but also innovatively embeds QC as a core decision-making engine. This establishes a closed-loop process of "perception-quantum optimization-decision-feedback," thereby breaking through the limitations of conventional DT systems in analytical and decision-making capabilities.
(2) A quantum modeling methodology for the reactive voltage power optimization (RVPO) problem in distribution networks is developed. This approach accurately transforms complex engineering optimization problems, which involve discrete/continuous variables, power flow constraints, and operational limits on device switching, into a quadratic unconstrained binary optimization (QUBO) model. It thereby addresses a key challenge in applying QC to practical engineering applications.
(3) A highly efficient and practical hybrid quantum-classical algorithm (HQCA) workflow is designed. This workflow leverages the stability of classical computing in model preprocessing, constraint handling, and post-processing, while synergistically utilizing the parallel search potential of QC for combinatorial optimization problems. It effectively mitigates the current limitations of quantum hardware in terms of qubit count and noise to offer a viable technical pathway toward achieving near-term quantum advantage.

II. THE RELATED WORK

A. Overall analysis and solution of the RVPO problem

Reactive Voltage Power Optimization (RVPO) in distribution systems is recognized as one of the core mechanisms for ensuring the secure, stable, and economical operation of power systems. It is essentially a complex combinatorial optimization problem characterized by multiple objectives, multiple constraints, non-convexity, and nonlinearity.[16] The integration of Distributed Renewable Energy (DRE) has further intensified the complexity of this problem.[17] The stochastic and fluctuating nature of DRE output leads to bidirectional and variable power flows, which significantly increase the risk of voltage violations.[18] A schematic representation of the multi-objective optimization framework for RVPO in distribution systems is illustrated in Fig. 1. Traditional optimization

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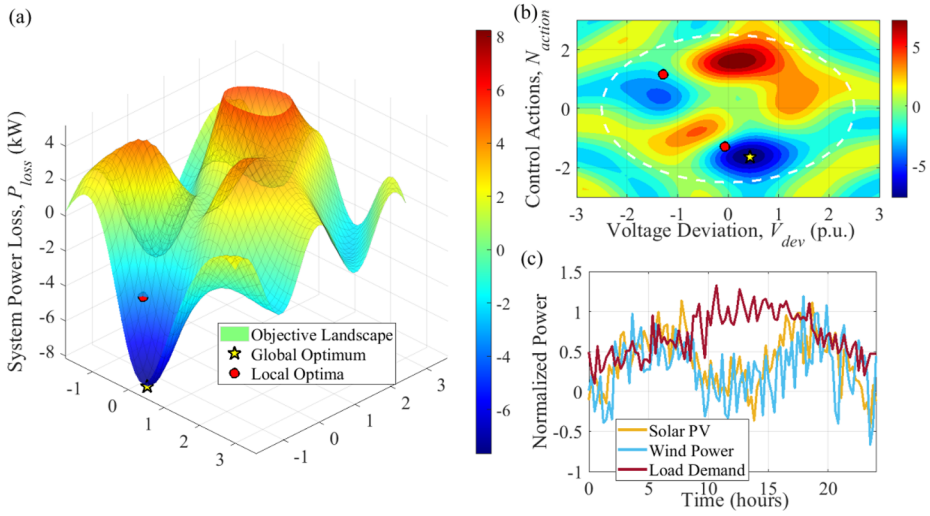


FIG. 1. Schematic diagram of the multi-objective optimization framework for RVPO in a distribution system. (a) System level: illustrates the multi-objective optimization framework focusing on minimizing network loss and voltage deviation. (b) Resource level: depicts the integration of stochastic and intermittent distributed PV and wind power. (c) Problem characteristic level: demonstrates the potential multiple local optima in a non-convex optimization problem.

objectives usually include the minimization of system loss, the minimization of voltage deviation, and the minimization of the number of actions of control devices. There is often a competitive relationship between these goals, which needs to be coordinated and balanced. The constraints include the equality constraints of power flow equations, the upper and lower limits of generator reactive power, the constraints of node voltage security, the constraints of OLTC (On-line Load Tap Changer) tap position, and the constraints of the number of switching capacitor banks. The mixed existence of discrete variables (such as OLTC position and switching state of capacitor banks) and continuous variables (such as node voltage and reactive power output) makes the problem difficult to be solved efficiently by conventional convex optimization methods.

Figure 1(a): A core method to ensure the safe, stable, and economical operation of power systems by managing reactive power and voltage levels.

Figure 1(b): Renewable energy sources (e.g., solar PV and wind) connected to the distribution grid, characterized by stochasticity and intermittency.

Figure 1(c): An optimization problem where the objective function or feasible region is not convex, often containing multiple local optima.

To solve the above-mentioned complexity problem, the existing research is mainly carried out along two paths.

(1) Improve the classical optimization algorithm.[19] For example, the interior point method is used to deal with the continuous variable optimization, or the intelligent algorithm [GA (Genetic Algorithm), PSO] is adopted to deal with the MINLP (Mixed-Integer Nonlinear Programming) problem.[20]

(2) An approximate reduction strategy, such as linearization or SOCR (Second-Order Cone Relaxation), is taken to transform the original non-convex problem into a tractable convex problem.[21]

B. Comparison and analysis of mainstream technical performance

In the simulation of a complex environment of a power system, there is often a contradiction between computational efficiency and global optimality, which has a risk of model distortion.[22] These methods are facing severe challenges when dealing with large-scale distribution network real-time optimization problems, as shown in Tables I and II.

Faced with the limitations of a single technology, it cannot fully meet the system requirements. Therefore, it is necessary to combine the complementary optimization of multiple methods to ensure the optimal state of the system.

TABLE I. Comparison of advantages and disadvantages of mainstream optimization techniques.

Technology category	Advantages	Disadvantages
Mathematical programming method[23]	The mathematical model is rigorous, the convergence speed is fast, and the precision is high	The ability to deal with non-convex problems is weak, which affects the accuracy of the solution
Swarm intelligence algorithm[24]	It does not depend on the objective function and deals with complex constraints easily	A large number of population iterations are needed, and the convergence speed is slow
QC[25]	Using quantum superposition, it is suitable for dealing with explosion	The algorithm needs to be transformed into a specific form, which may introduce errors

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TABLE II. Comparison of technology combinations and complementarity.

Technology combination mode	Combined description	Complementarity
Classical optimization algorithm + heuristic rule[26]	The mathematical programming method is used to solve the sub-problems of continuous variables and deal with discrete variables	It improves the computational efficiency and is difficult to ensure the overall optimality
Intelligent algorithm + local search[27]	Intelligent algorithms are used for global exploration, and local search methods are adopted for fine development	The convergence accuracy is improved, but the computational cost is further increased
DT+HQCA	Using DT to decompose the original problem, the optimization part is solved, and the rest is handled by HQCA	The potential of DT in complex combinatorial problems is to make up for the lack of quantum hardware by using the generality and stability of HQCA

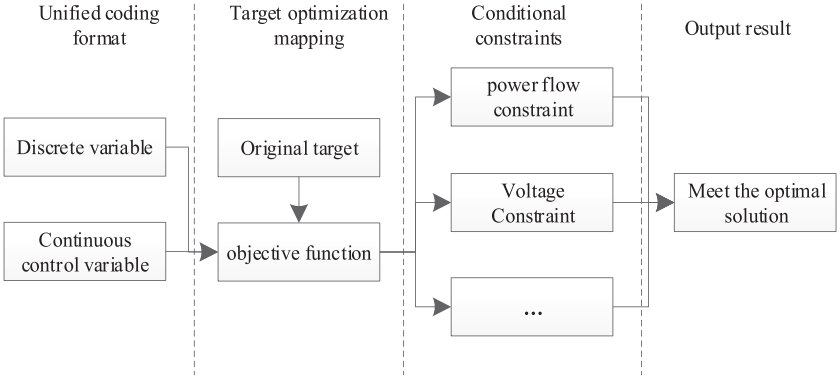


FIG. 2. The quantum unconstrained transformation process.

Digital twins provide high-fidelity models and real-time data, while quantum algorithms are responsible for core optimization. Through the integration of these technologies, breakthroughs in computational speed and solution accuracy can be achieved.

Based on the above-mentioned analysis, this paper adopts a Hybrid Quantum-Classical Algorithm (HQCA) framework to address the RVPO problem. The core technical approach involves transforming the original Mixed-Integer Nonlinear Programming (MINLP) problem into a Quadratic Unconstrained Binary Optimization (QUBO) model, which is compatible with quantum computing paradigms. The QUBO formulation serves as a standard input for quantum annealers and certain quantum circuit algorithms, such as the Quantum Approximate Optimization Algorithm (QAOA). It is mathematically represented as follows:

$$\text{minimize} H(\mathbf{x}) = \mathbf{x}^T \mathbf{Q}\mathbf{x} + c = \sum_i \sum_j Q_{ij} x_i x_j + c, \tag{1}$$

where $H(\mathbf{x})$ represents the objective function value to be minimized; $\mathbf{x}$ represents the binary decision vector, $x_i, x_j \in \{0, 1\}$; and $\mathbf{Q}$ represents a real symmetric square matrix. The diagonal elements Q_{ii} are the linear coefficients of the x_i variables, the off-diagonal element Q_{ij} is the quadratic coupling coefficients between variables x_i and x_j; and c is a constant term.[28]

Figure 2 shows the transformation process from the RVPO problem to the QUBO model for quantum unconstrained optimization.

(1) All discrete and continuous control variables shall be coded in a specific way (binary coding);
(2) Map the original optimization objective (such as minimizing the network loss) to the objective function $H(\mathbf{x})$ of the QUBO model;
(3) Introduce all constraint conditions (such as power flow constraint and voltage constraint) into the objective function as a penalty term, wherein the mathematical expression is as follows:

$$H_{\text{total}} = H_{\text{objective}} + \lambda P_{constraints}, \tag{2}$$

where, H_{total} is the total objective function of QUBO after conversion, $H_{\text{objective}}$ is the part converted from the original objective function, $P_{\text{constraints}}$ is the penalty function to measure the degree of constraint violation, and λ is a sufficiently large penalty coefficient to ensure that the constraint conditions are satisfied at the optimal solution.[29]

III. CONSTRUCTION AND MODELING OF QUANTUM HYBRID ENHANCED DT SYSTEM

In this study, the new energy grid-connected distribution system QC-enhanced DTS is studied, and an intelligent platform connecting the physical distribution network and virtual space is constructed.

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A. Research object

The physical object is an active distribution network connected to a high proportion of distributed photovoltaic and wind power. The system contains a variety of controllable resources, such as shunt capacitor banks, on-load tap changers (OLTCs), SVCs (Static Var Compensators), and inverter-based distributed generators with reactive power support capability. The coordinated control of this equipment is key to maintaining the voltage stability of the system. The DT layer synchronizes the state of physical entities in real time through the Internet of Things technology and constructs a high-fidelity simulation model. The quantum enhancement layer is the technical core of this paper, and its task is to transform the traditional RVPO problem into a form that can be handled by QC and solve it efficiently through HQCA. Finally, the optimal strategy is fed back to the physical system for execution, forming a "perception-decision-control" closed-loop.

B. Technology application and innovation

The technical application of this study is reflected in the integration of QC as a service into the decision engine of DT. DT technology is responsible for multi-source data fusion, high-precision simulation, and visualization of results, providing real scenarios and constraints for optimization. QC is applied to solve the core combinatorial optimization problems that are computationally expensive in traditional methods, such as the joint optimization of OLTC gear and capacitor switching state. The innovation lies in the fact that it does not rely entirely on quantum hardware but designs a collaborative computing framework. In this framework, the processing flow is responsible for pre-processing (such as scenario generation and problem size reduction), post-processing (such as interpretation and feasibility verification), and interaction with the DT model. The quantum coprocessor focuses on solving the transformed QUBO model, giving full play to its parallel search advantages. This architecture effectively circumvents the limitations of quantum hardware in the current NISQ (Noisy Intermediate-Scale Quantum) era and provides a feasible path to achieve practical quantum advantages in the near future.

C. System technical framework and modeling

The QC-enhanced DTS technical process proposed in this study follows a closed-loop optimization process, as shown in Fig. 3.

Figure 3 illustrates the architecture of the distribution network physical system, which comprises three layers: the Digital Twin (DT) layer, the quantum service layer, and the classical computing layer. First, the system collects the real-time operation data of the physical distribution network through intelligent sensors and measurement devices. The data include node voltage, branch power, load level, and the output of DERs (Distributed Energy Resources). The data are transmitted to the DT layer through the communication network. In the DT layer, a high-fidelity multi-physics simulation model is constructed and continuously updated. The model accurately reflects the topology, device parameters, and real-time status of the physical network. Based on this model, a system state estimation program is run to obtain a complete and consistent system operating profile. This section, together with the predicted new energy output and load data, is input to the optimization decision module.

The optimization decision module is the core of the intersection of QC and classical computing. Here, the processing flow models the RVPO problem mathematically as described in Sec. III D and transforms it into the QUBO form.

A set of candidate solutions is returned to the process flow for post-processing. The post-processing includes decoding the binary solution to the actual control variables (such as gear and switching state), checking the feasibility of the power flow calculation, and finally calculating the objective function value. The optimal control strategy is screened out and sent to the actuator of the physical system after safety verification to complete an optimal control cycle.

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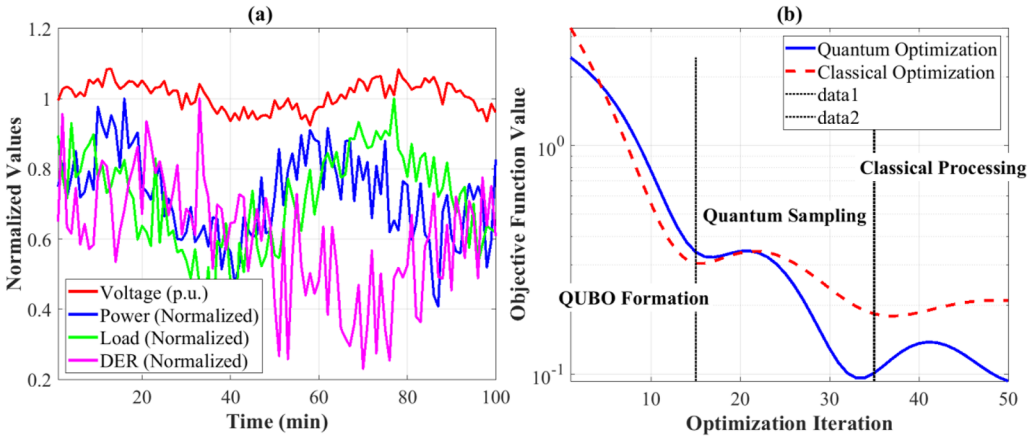


FIG. 3. Architecture of the quantum computing-enhanced digital twin system for reactive power-voltage optimization. (a) Physical distribution network layer: comprises smart sensors, controllable devices (OLTC, SVC, and capacitor banks), and distributed energy resources (PV and wind power) for real-time data acquisition and control execution. (b) Quantum service layer: responsible for transforming the RVPO problem into a QUBO model and executing quantum optimization algorithms to search for global optimal solutions.

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D. Mathematical model and formula derivation

The classical mathematical model of the RVPO problem usually takes the minimization of the total active power loss of the system as the primary objective, while the voltage deviation is taken as the secondary objective or constraint. The objective function can be expressed as

$$\min F = P_{\text{loss}} + \lambda_v \sum_{i \in N} (V_i - V_{\text{ref}})^2, \tag{3}$$

where F denotes the comprehensive optimization objective; P_{loss} denotes the total active power loss of the system; λ_v denotes the penalty coefficient of the voltage deviation item, which is a large positive number; V_i is the voltage amplitude of node i; V_{ref} is the voltage reference value, usually 1.0 (p.u); and N denotes the set of all nodes of the system.[30]

The objective function should satisfy a series of equality and inequality constraints. The core constraint is the power flow balance equation, namely,

$$\begin{aligned} P_{Gi} - P_{Li} &= V_i \sum_{j=1}^{n} V_j (G_{ij} \cos \theta_{ij} + B_{ij} \sin \theta_{ij}), \\ Q_{Gi} - Q_{Li} &= V_i \sum_{j=1}^{n} V_j (G_{ij} \sin \theta_{ij} - B_{ij} \cos \theta_{ij}). \end{aligned} \tag{4}$$

Here, P_{Gi} and Q_{Gi} are, respectively, the active and reactive outputs of the generator connected to node i; P_{Li} and Q_{Li} are, respectively, the active and reactive power of the load at node i; G_{ij} and B_{ij} are, respectively, the real and imaginary parts of the elements of the node admittance matrix; and θ_{ij} represents the voltage phase angle difference between node i and node j.

The problem also contains the following inequality constraints:

$$\begin{aligned} &V_i^{\min} \le V_i \le V_i^{\max}, \forall i \in N, \\ &Q_{Gk}^{\min} \le Q_{Gk} \le Q_{Gk}^{\max}, \forall k \in G, \\ &T_m^{\min} \le T_m \le T_m^{\max}, \forall m \in T, \\ &C_n \in \{0,1\}, \forall n \in C. \end{aligned} \tag{5}$$

Here, $V_i^{\min}$ and $V_i^{\max}$ are, respectively, the upper and lower safety limits of the node voltage; $Q_{Gk}^{\min}$ and $Q_{Gk}^{\max}$ are, respectively, the upper and lower limits of the output of the reactive power source k; $T_m^{\min}$ and $T_m^{\max}$ are, respectively, the gear limit of the OLTC (On-Load Tap Changer) tap m; Cn represents the switching state of the capacitor bank n and is a binary variable; G, T, and C are, respectively, the set of the reactive power source, OLTC, and capacitor bank.

In order to transform the MINLP problem into the QUBO form, continuous variables and constraints are treated. First, continuous variables such as reactive power outputs and voltage amplitudes need to be discretized. Second, all constraints must be added to the objective function as penalty terms.

For an inequality constraint $x \le b$, it can be transformed into an equality constraint $x + s = bx + s = b$ and a penalty term $\lambda(x + s - b)^2$ can be introduced, where s is a slack variable.

After discretization and penalty treatment, the original problem can be reformulated as an unconstrained optimization problem suitable for QC.[31] Its final form is to find a binary vector $\mathbf{u}_i$ to minimize the following Hamiltonian:

$$H(\mathbf{u}_i) = \mathbf{x}^T \mathbf{Q} \mathbf{x}. \tag{6}$$

Here, $\mathbf{x}$ represents a vector composed of all binary decision variables (such as equipment status and discretized reactive power output level), and $\mathbf{Q}$ represents a symmetric coefficient matrix, whose elements encode the cost of the original objective function and the penalty weights of all constraints.

The core innovation is the construction of a matrix $\mathbf{Q}$ whose diagonal elements Q_{ii} and variable x_i vary their own energy offsets or costs. The off-diagonal elements Q_{ij} represent the coupling strength between the variables x_i and x_j. When the simultaneous action of two reactive power compensation devices may cause the voltage to exceed the limit, the corresponding coupling term will have a large positive penalty coefficient to avoid this situation in the optimization.

Algorithm pseudo code:
Algorithm: HQC-RVO Algorithm
Input: $\mathbf{Y}t$, $\mathbf{F}t + 1$
Output: $\mathbf{u}^*$ Preprocess:
1. $\min_{\mathbf{u}} f(\mathbf{x}, \mathbf{u})$ s.t. $\mathbf{g}(\mathbf{x}, \mathbf{u}) = 0, \mathbf{h}(\mathbf{x}, \mathbf{u}) \le 0$
2. $\mathbf{u} \to \mathbf{z} \in 0, 1^n$
3. Formulate QUBO:
4. $H(\mathbf{z}) = \mathbf{z}^T \mathbf{Q} \mathbf{z} + \mathbf{c}^T \mathbf{z}$
5. Quantum Sampling:
6. $X_{\text{cand}} \leftarrow \text{QPU}(\mathbf{Q}, \mathbf{c})$
7. Postprocess:
8. $\mathcal{F} \leftarrow \oslash$
9. for $\mathbf{z}i \in X$cand do
10. $\mathbf{u}_i \leftarrow \text{decode}(\mathbf{z}_i)$
11. if feasible$(\mathbf{u}_i)$ then $\mathcal{F} \leftarrow \mathcal{F} \cup \mathbf{u}_i$
12. end for
13. Select:
14. $\mathbf{u}^* \leftarrow \arg\min_{\mathbf{u} \in \mathcal{F}} f(\mathbf{x}, \mathbf{u})$
15. Return: $\mathbf{u}^*$
16. End

By designing a matrix in this study, complex engineering optimization problems are successfully mapped into the solvable space of QC. The key innovation lies in the design of the hybrid computing workflow of the system and the in-depth derivation of the mathematical process of transforming the traditional MINLP problem into the QUBO model that can be handled by QC. By variable discretization and constraint penalization, the transformation is realized to lay a theoretical foundation for an efficient solution by a quantum algorithm.

IV. DESIGN OF THE QUANTUM-CLASSICAL HYBRID COOPERATIVE OPTIMIZATION ALGORITHM

According to the quantum bit resource mapping model proposed in this study, a hybrid coding scheme based on PLA (Piecewise Linear Approximation) is adopted.

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A. Qubit mapping and coding model

The advantage of this encoding scheme lies in its significant reduction of quantum bit resources, thereby enabling the application to practical-scale distribution network problems on near-term Noisy Intermediate-Scale Quantum (NISQ) processors.[32]

Continuous control variables, such as the reactive power output Q_{svc} of a Static Var Compensator (SVC), are first discretized into multiple intervals. Instead of employing standard binary encoding with multiple binary variables for each interval, a combined method of one-hot encoding and amplitude encoding is adopted.[33] Suppose the reactive power adjustment range of an SVC is discretized into K levels. Then K qubits denoted as $q_1, q_2, \ldots, q_k$ are used for one-hot encoding.

In addition, M extra qubits are utilized to encode fine deviations within each interval using binary amplitude encoding. The corresponding reactive power output can be expressed as

$$Q_{svc} = \sum_{K}^{k=1} \delta_k \cdot \left(Q_k^{\min} + \frac{Q_k^{\max} - Q_k^{\min}}{2^M - 1} \sum_{M}^{m=1} 2^{m-1} q_{k,m} \right), \tag{7}$$

where, Q_{svc} is the actual reactive power output value of SVC; δ_k is a binary indicator variable, which is 1 if and only if the output is in the k th interval, and is determined by the one-hot coding qubit; $Q_k^{\min}$ and $Q_k^{\max}$, respectively, define the lower limit and upper limit of the k th interval; M represents the number of qubits used for amplitude coding; and $q_{k,m} \in \{0, 1\}$ represents the qubit state of the m th amplitude bit in the k th interval.

B. Constraint satisfaction quantum Hamiltonian construction

In the QAOA, the construction of the problem Hamiltonian H_C is crucial.[34] In this paper, all the hard constraints and soft targets are encoded into the Hamiltonian in an innovative way. For inequality constraints, such as voltage security constraints, a quadratic penalty method based on auxiliary slack bits is used.

Take the node voltage out-of-limit constraint $V_i^{\min} \le V_i \le V_i^{\max}$ as an example.

(1) This constraint is translated into two inequalities,

$$V_i - V_i^{\max} \le 0, V_i^{\min} - V_i \le 0. \tag{8}$$

A non-negative slack variable s is introduced for each inequality, which is discretized and encoded with additional qubits.

(2) The penalty Hamiltonian term corresponding to the constraint is

$$H_{\text{volt}} = \lambda_v \sum_{i \in N} \left[\left(V_i(\mathbf{x}) - V_i^{\max} + s_{i1} \right)^2 + \left(V_i^{\min} - V_i(\mathbf{x}) + s_{i2} \right)^2 \right]. \tag{9}$$

Here, H_{volt} is the Hamiltonian of the penalty term of the voltage out-of-limit constraint; λ_v is a sufficiently large penalty coefficient to ensure that the constraint is satisfied at the optimal solution; N is the set of system nodes; $V_i(\mathbf{x})$ is the voltage amplitude of node i and the function of the decision variable encoding vector $\mathbf{x}$, which is obtained by forward and backward power flow calculation or linearization of the power flow model; and s_{i1} and s_{i2} are the discretized slack variables introduced by the two inequality constraints at the i th node, encoded by the extra qubits.

(3) The problem Hamiltonian Hc consists of the objective function term and all the constraint penalty terms,

$$H_C = H_{\text{loss}} + H_{\text{volt}}. \tag{10}$$

Here, H_{loss} represents the Hamiltonian corresponding to the system network loss target item and H_{power} represents the Hamiltonian of the penalty term corresponding to the power balance equality constraint.

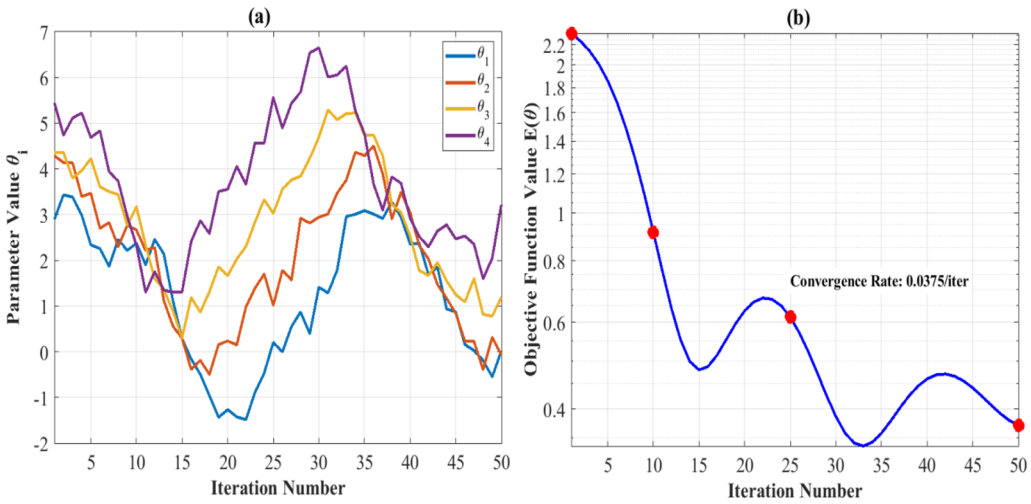


FIG. 4. Flowchart of the variational quantum-classical hybrid solving algorithm. (a) Parameter initialization: generates initial values for variational parameters to initialize the quantum circuit. (b) Expectation value calculation: measures the quantum state to compute the expected value of the problem Hamiltonian, which is fed back to the classical optimizer.

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The construction method ensures that the lowest energy state (ground state) corresponds to the optimal solution satisfying all constraints.

C. Variational quantum-classical hybrid solution process

In this study, a variational quantum-classical hybrid solution process is designed, and its core framework is shown in Fig. 4. The process efficiently decomposes the optimization task between the quantum processor and the classical processor.

Figure 4 illustrates the cyclic process of parameter initialization, quantum state preparation, expectation value calculation, and parameter updated by the classical optimizer.

The algorithm starts with a set of randomly initialized variational parameters θ. On a quantum processor, the preparer prepares a quantum state $|\psi(\theta)\rangle$ through a set of quantum gate operations controlled by these parameters.

The expected value $\langle H_C \rangle$ is returned to the process flow, and the updated parameters are sent to the quantum processor again to start a new iteration. This loop continues until convergence or a maximum number of iterations is reached.

The mathematical objective of the process can be stated as

$$\min_{\theta} \langle \psi(\theta) | H_C | \psi(\theta) \rangle. \tag{11}$$

Here, θ denotes the variational parameter vector, which usually corresponds to a set of rotation angles in QAOA, $|\psi(\theta)\rangle$ denotes the quantum state prepared by the parameterized quantum circuit, and Hc is the constructed problem Hamiltonian.

The hybrid framework gives full play to their respective advantages by transferring the heavy computational burden of quantum state preparation and expectation value calculation to quantum devices, while the complex parameter optimization is completed by efficient classical optimizers.

D. Model verification and error compensation mechanism

Due to the errors introduced by the noise and model simplification in the QC process, a closed-loop verification and compensation mechanism is integrated into the algorithm. After the quantum optimizer gives the candidate solution, the solution is substituted into DT for accurate high-fidelity power flow calculation. The true objective function value of F_{true} is calculated, and the constraint violation degree is evaluated.

The optimization error is defined as

$$\epsilon = |F_{\text{true}} - \langle H_C \rangle|. \tag{12}$$

Here, ϵ denotes the discrepancy between the quantum optimization result and the true value, F_{true} represents the ground-truth objective function value obtained via accurate power flow validation within the Digital Twin (DT), and $\langle H_C \rangle$ corresponds to the expected value of the objective function derived during the quantum optimization process.

If the error or constraint violation exceeds a predetermined threshold, a compensation mechanism will be initiated. The compensation mechanism can be a fast classical local search with the quantum solution as the starting point for fine-tuning, or it can be to adjust the penalty coefficient and re-perform the quantum optimization. This mechanism ensures that the final solution not only has the global exploration advantage of QC but also meets the actual engineering accuracy requirements.

This study focuses on quantum-classical hybrid collaborative optimization algorithms and models for RVPO problems. The main innovations are as follows: First, a hybrid coding scheme is proposed to save quantum bit resources and efficiently deal with continuous-discrete hybrid variables. Second, the construction method of the Hamiltonian encoding hard constraints and soft targets is designed. Furthermore, the variational quantum-classical hybrid solution process is planned to make full use of the two types of computing resources. In addition, based on sensitivity analysis, a preprocessing strategy is introduced to reduce the problem size. Finally, an error compensation mechanism integrated with DT verification is designed to ensure the feasibility of the solution. These models and algorithms together constitute the core technology path to achieve the advantages of QC enhancement.

V. SIMULATION EXPERIMENT ANALYSIS

The effectiveness and superiority of QC-enhanced DTS and HQCA proposed in this study in solving the RVPO problem of a high proportion of new energy distribution networks are verified. By evaluating the test items, such as the optimization effect of the proposed method, the quantitative basis for the application of QC in power system optimization is provided.

A. Experimental simulation environment

The simulation platform integrates classic power system simulation tools and the QC simulator. The platform configurations are shown in Tables III–V.

TABLE III. Hardware configuration of the simulation platform.

Items	Parameters	Functions
Server	2x Intel Xeon Gold 6248R, 512GB RAM	Run the DT model, quantum emulator
QC simulator	NVIDIA A100 80GB PCIe	Simulate noiseless quantum processing, perform QAOA algorithm
Data storage	4TB NVMe SSD RAID	Storage system topology and operation log

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TABLE IV. Software configuration of the simulation platform.

Items	Parameters	Functions
DT platform	MATLAB/Simulink R2023a, OpenDSS	Construction of distribution network high-fidelity model, power flow calculation and simulation
Classical optimization solver	IBM ILOG CPLEX optimization Studio 22.1.1	MIP (mixed-integer programming)
A framework for quantum algorithms	Qiskit 0.43.0, Pennylane 0.32.0	Parameterized quantum circuits are constructed to implement the QAOA algorithm
Development language	Python 3.10	Integrate various platforms to realize algorithm flow and control logic

TABLE V. Training parameters for the QAOA.

Items	Parameters	Functions
Number of algorithm layers (p)	3	Control QAOA circuit depth
Classic optimizer	COBYLA	Optimizing the variational parameters of quantum circuits
Maximum number of iterations	200	Limit the maximum number of runs of the classical optimizer
Penalty factor (λ)	1×10^5	Weights for objective function minimization

B. Simulation experiment analysis

Simulation is conducted on a modified IEEE 33-node distribution system (sample size: one primary system integrated with three distributed photovoltaic power stations and two wind farms). A total of 100 operational scenarios corresponding to different dates are generated to capture the variability of renewable generation and load demand. Each scenario undergoes a 24-h chronological simulation with a time resolution of 15 min.[35] The proposed HQCA is compared against two classical algorithms: the MIP solver CPLEX (Constraint Programming LEX)[36] and Particle Swarm Optimization (PSO). The Wilcoxon signed-rank test is employed to assess the statistical significance of performance differences at a significance level $\alpha = 0.05$.[37]

1. Comparison of optimization effect-total active power loss of the system

The validation aims to verify the effectiveness of the proposed HQCA in achieving the core optimization objective and to evaluate its performance relative to classical algorithms. Consequently, the efficacy is directly measured by the capability of different algorithms in reducing system operational costs. The test results are presented in Table VI.

The results indicate that the HQCA yielded the lowest average power loss of 76.42 kW. A Wilcoxon signed-rank test was conducted (sample size: 100 scenarios; significance level $\alpha = 0.05$), revealing that the performance of the HQCA is significantly superior to that of PSO ($p < 0.01$). In contrast, no statistically significant difference was observed between the HQCA and the MIP solver CPLEX ($p > 0.05$). These findings confirm that the HQCA achieves optimization quality comparable to that of state-of-the-art classical optimizers, while also laying the groundwork for potential improvements in computational efficiency on future quantum hardware. This outcome validates the advanced capability and practical utility of the proposed approach.

2. Voltage quality improvement comparison

The ability of HQCA is tested and evaluated to improve the voltage quality of the distribution system.

TABLE VI. Comparison of average total active power loss (unit: kW).

Algorithms	Average value	Standard deviation	Minimum value	Maximum value
HQCA (in the study)	76.42	5.18	68.91	87.54
MIP (CPLEX)	76.85	5.31	69.33	88.17
PSO	78.93	5.67	71.05	90.26
(Sample size: $n = 100$ running scenarios)				

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The calculation formula is as follows:

$$\Delta V_{\text{avg}} = \frac{1}{N \cdot T} \sum_{t=1}^{T} \sum_{i=1}^{N} |V_{i,t} - V_{\text{ref}}|. \quad (13)$$

Here, ΔV_{avg} is the average voltage deviation, N is the total number of system nodes, T denotes the total number of time points, $V_{i,t}$ denotes the voltage amplitude of node i at time t, and V_{ref} is the voltage reference value, which is 1.0 (p.u).

Voltage stability is a key indicator of the safe operation of the system. The test verifies the control efficacy of different optimization algorithms on the voltage level by quantifying the average voltage deviation. The results are shown in Table VII.

The results demonstrate that the HQCA significantly reduces the average voltage deviation to 0.0087 p.u., outperforming both PSO (0.0095 p.u.) and MIP (0.0091 p.u.) under a sample size of 100 scenarios. The statistical significance of these differences was confirmed using the Wilcoxon signed-rank test at α = 0.05. Further analysis indicates that the HQCA excels in coordinating multiple types of reactive power resources, thereby maintaining voltages closer to the reference values more effectively. This capability underscores its advanced performance and effectiveness in enhancing system voltage security and power quality, as shown in Fig. 5–7.

3. Robustness test for dealing with uncertainty

To evaluate the robustness of the HQCA under uncertainties involving prediction errors, the performance degradation of each algorithm's optimized solution under actual power fluctuations was assessed. A robustness index, defined as the ratio of actual power

TABLE VII. Comparison of average voltage deviation (unit: p.u).

Algorithms	Average value	Standard deviation	Minimum value	Maximum value
HQCA (in the study)	0.0087	0.0012	0.0065	0.0113
MIP (CPLEX)	0.0091	0.0013	0.0069	0.0118
PSO	0.0095	0.0014	0.0072	0.0125
(Sample size: n = 100 running scenarios)				

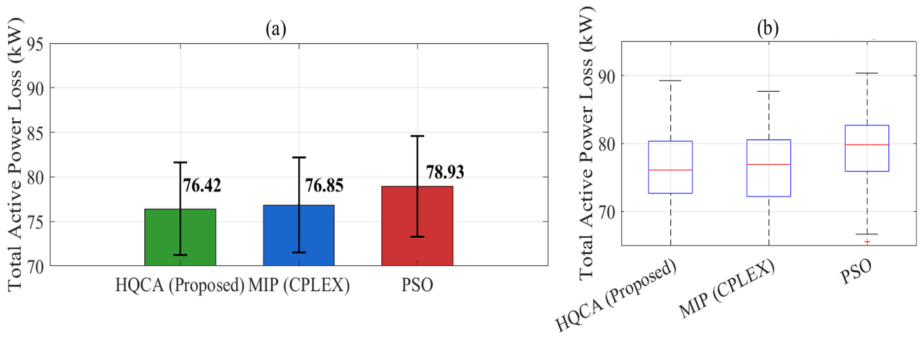


FIG. 5. Comparison of optimization effect-total active power loss of system. (a) Bar chart of average loss: visually compares the average total active power loss obtained by the HQCA, MIP (CPLEX), and PSO algorithms. (b) Box plot of statistical loss distribution: presents the statistical spread (including median and range) of loss values for each algorithm using synthesized data.

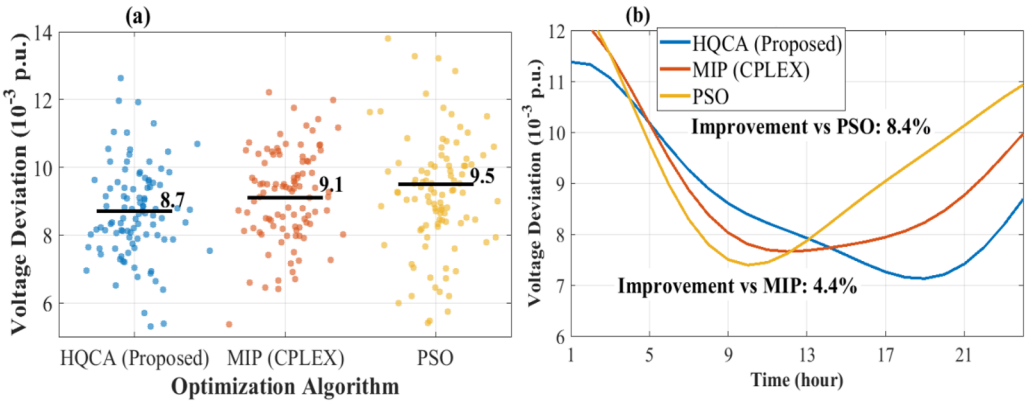


FIG. 6. Voltage quality improvement comparison. (a) Voltage deviation scatter distribution: displays the voltage deviation of each scenario for the three algorithms (HQCA, MIP, and PSO) in a scatter plot, reflecting the dispersion of optimization results. (b) 24-h voltage profile comparison: shows the variation in average node voltage deviation over a 24-h period, reflecting the algorithms' adaptability to time-varying loads and renewable energy output.

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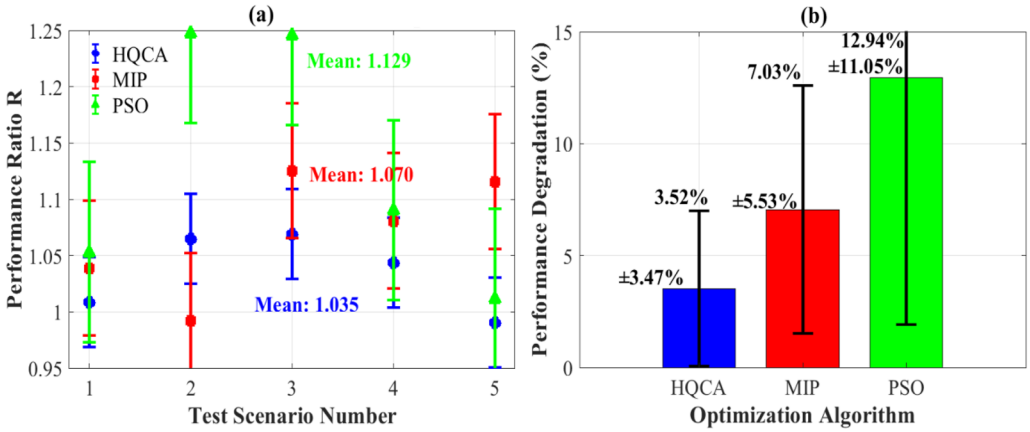


FIG. 7. Test and analysis of the robustness of uncertainty. (a) Bar chart of robustness performance comparison with error bars: compares the average performance ratios of the three algorithms under five extreme volatility scenarios. (b) Chart of average performance degradation percentage: calculates the average percentage of performance degradation caused by uncertainty for each algorithm; lower values indicate stronger robustness (sample size: n = 100 optimization solutions; error bars are 95% confidence intervals).

TABLE VIII. Comparison of robustness performance (ratio of actual loss to predicted loss).

Algorithms	Average value	Standard deviation	Minimum value	Maximum value
HQCA (in the study)	1.03	0.04	0.98	1.08
MIP (CPLEX)	1.07	0.06	1.00	1.14
PSO	1.10	0.08	1.02	1.20
(Sample size: n = 5 extreme volatility scenarios)				

loss to the ideal predicted loss, was introduced to directly reflect an algorithm's ability to cope with uncertainty. A value closer to 1 indicates stronger robustness. The corresponding data analysis is summarized in Table VIII.

The results demonstrate that the HQCA scheme achieved an average ratio of 1.03, corresponding to a performance degradation of only 3%. This performance is superior to that of the MIP (1.07) and PSO (1.10) methods (sample size: n = 5 extreme volatility scenarios; statistical method: arithmetic mean calculation).

Further analysis reveals that the global search capability of the HQCA enables it to identify solutions situated in flatter regions of the optimization landscape, which exhibit lower sensitivity to parameter variations. This property allows the algorithm to maintain robust performance under uncertain conditions, which validates the effectiveness and practical utility of the proposed approach in handling real-world system uncertainties.

4. Convergence analysis of the algorithm

An in-depth analysis of the convergence behavior of the variational quantum optimization process within the HQCA was conducted. The high proportion of renewable energy connected to the grid leads to a significant increase in the uncertainty of prediction, and the optimization algorithm needs to maintain stable performance under this condition. By introducing extreme fluctuation scenarios, the test evaluates the performance attenuation degree of

TABLE IX. Analysis of convergence characteristics for the HQCA algorithm.

Convergence phase	Range of iterations	Mean descent gradient (iteration)	Percent reduction in function values (%)
Period of rapid decline	1–40	0.025	68.5
Stable optimization period	41–100	0.008	24.3
Stable period of convergence	101–120	0.002	7.2
(Sample size: n = 1 typical scenario; convergence threshold: <0.1% change over 10 consecutive iterations)			

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each algorithm optimization scheme under real fluctuation. The convergence characteristics serve as key metrics for assessing algorithmic stability and computational feasibility, directly influencing the reliability and reproducibility of the optimization outcomes. The corresponding results are summarized in Table IX and illustrated in Table IX.

Figure 8 shows the small fluctuations of the objective function in the convergence region and the final stable state. Besides, the threshold value indicates the level of the objective function when the algorithm reaches the stable state (<0.1% change in 10 consecutive iterations).

Figure 9 shows the distribution of objective function values in the two-dimensional space formed by parameters α and β. The optimization trajectory of the algorithm in the parameter space gradually converges to the mapping relationship of the objective function value in the optimal solution area from the initial point.

The experimental results verify that the algorithm exhibits favorable convergence properties, with the objective function value decreasing steadily as the number of iterations increases, eventually reaching a stable plateau after ~120 iterations (sample size: n = 1 typical scenario; performance metric: normalized expected objective function value).

Further analysis indicates that this convergence trajectory confirms the capability of the parameterized quantum circuit to effectively explore the solution space and progressively approach the region containing the optimal solution. A synergistic interplay

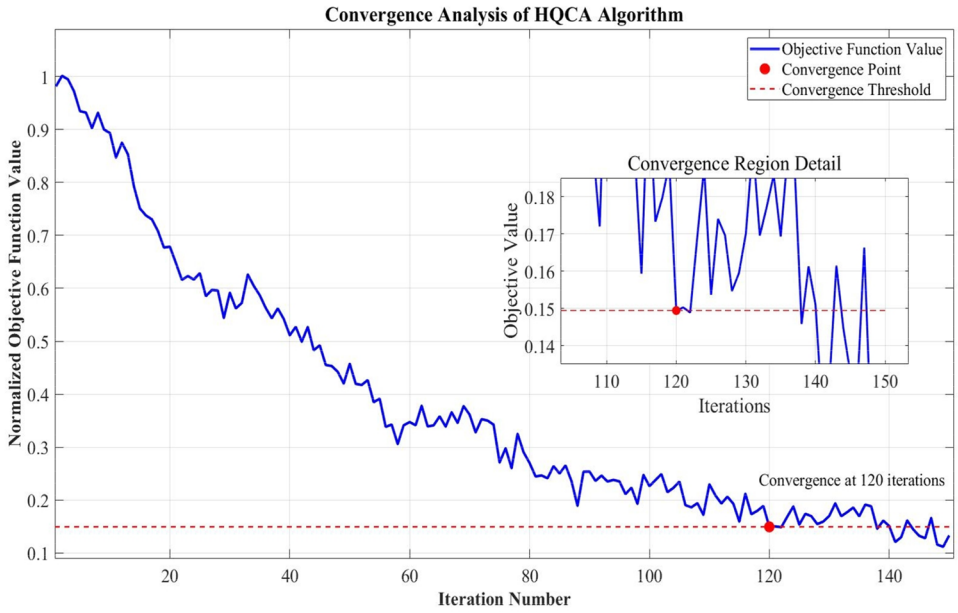


FIG. 8. Convergence curve of the HQCA algorithm.

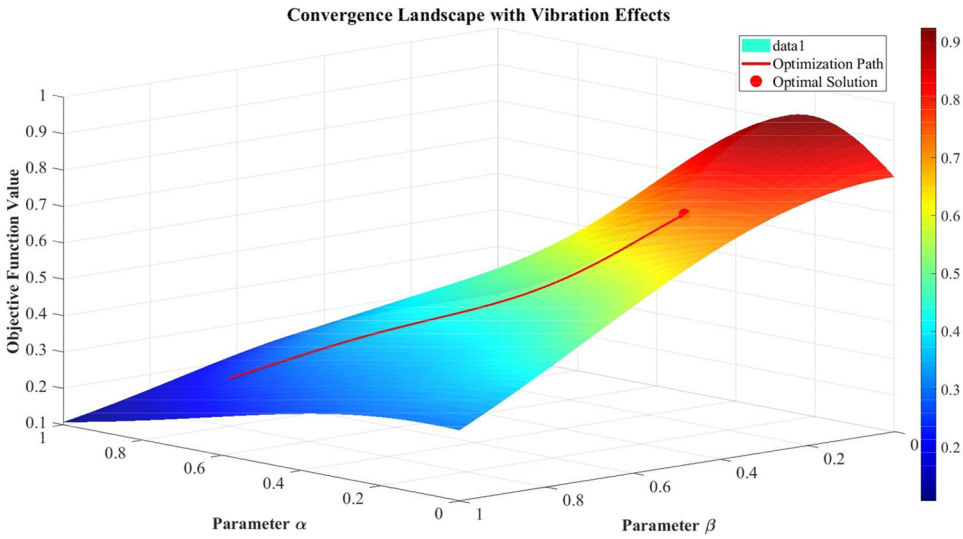


FIG. 9. Convergence landscape with vibration effects.

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between the classical optimizer and quantum expectation estimation has been established. Although there is still room for improvement in the convergence rate, the stable convergence behavior confirms the feasibility and reliability of the proposed HQCA framework. Thus, it provides an important basis for the numerical stability of the algorithm in practical engineering applications and verifies the advancement of the proposed method in the realization of quantum-classical collaborative optimization.

C. Discussion

Simulation results confirm the effectiveness of the proposed hybrid quantum-classical optimization framework. The framework achieves optimization quality comparable to that of traditional mixed-integer programming (MIP) solvers, while demonstrating superior performance in terms of voltage quality and robustness. Although the computational efficiency does not exhibit a definitive advantage in the current simulation environment, the underlying computational paradigm suggests potential for accelerated performance when addressing larger-scale problems or when executed on actual quantum hardware. However, this study has certain limitations: it does not fully capture the noise characteristics of real quantum devices and their impact on algorithmic performance; furthermore, the scalability of the algorithm necessitates further validation on more complex and practical grid systems beyond the distribution network test case used herein.

VI. CONCLUSION

By devising a quantum-classical hybrid collaborative optimization algorithm and model, this paper addresses the reactive power-voltage optimization problem in distribution networks. The main contributions are summarized as follows:

(1) A hybrid encoding scheme is proposed to conserve qubit resources and efficiently handle mixed continuous and discrete variables.
(2) A Hamiltonian construction method is designed to incorporate both hard constraints and soft objectives within a unified encoding framework.
(3) A variational quantum-classical hybrid solution workflow is developed to leverage the computational strengths of both quantum and classical resources.
(4) Based on sensitivity analysis, a preprocessing strategy is introduced to reduce problem dimensionality.

The experimental analysis demonstrates that the integrated digital twin-based error compensation mechanism ensures solution feasibility. Collectively, these models and algorithms form a core technical pathway toward achieving quantum-enhanced computational advantages.

Building on these findings, future work will study validating the algorithm's performance on real quantum hardware and assessing the model's robustness under complex operating conditions. In addition, the system will be scaled to a 110 kV grid level to evaluate its practical applicability. Further research will aim to enhance the efficiency of the quantum algorithmic framework, improve computational and optimization precision, and broaden the application scope of quantum computing in power systems.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

The authors Chenda Jiang and Jin Li contributed equally to this paper. All authors reviewed the manuscript.

Chenda Jiang: Conceptualization (equal); Data curation (equal); Formal analysis (equal); Methodology (equal); Writing – original draft (equal); Writing – review & editing (equal). **Jin Li**: Formal analysis (equal); Methodology (equal); Resources (equal); Software (equal); Writing – original draft (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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